

GradeLine
User Guide

Existing-ground long sections and driveway design from survey DXF data

Document	GradeLine User Guide
Applies to	GradeLine web, single-file and Windows desktop builds as at 31 October 2026
Prepared by	Thomas Renoden
Document date	31 October 2026
Project repository	https://github.com/TReno06/GradeLine
Hosted application	https://treno06.github.io/GradeLine/

Intended use. GradeLine is a screening and visualisation aid. It does not replace engineering design, a council's own requirements, or professional sign-off. Section 13 sets out its limits.

Contents

1. Introduction . . . 3
2. Installation and launch . . . 3
3. Loading survey data . . . 4
4. Preparing the data . . . 5
5. The views . . . 6
6. Reading the long section . . . 7
7. Driveway design . . . 8
8. Vehicle scrape check . . . 10
9. Crossfall comparison . . . 11
10. Saving and reopening jobs . . . 12
11. Exporting results . . . 13
12. Troubleshooting . . . 14
13. Limitations and assumptions . . . 15
Appendix A. Default values . . . 16
Appendix B. Glossary . . . 17

1. Introduction

GradeLine reads survey data from a DXF file, builds a triangulated ground surface (a TIN), cuts a long section along one or more alignments, and lets the user lay a straight-grade driveway design over that section and check it against AS 2890.1. It runs entirely on the user's own computer. No survey data is uploaded, and the application works offline.

1.1 Purpose

This guide explains how to load survey data, prepare it, read the resulting views, design and check a driveway, compare two alignments, save work, and export results to AutoCAD or PDF. It is written for civil and survey practitioners who are comfortable with chainages, reduced levels (RL) and long sections.

1.2 Capabilities

- Build a 3D surface from **POINT** and **3DFACE** survey shots, with optional contour polylines.
- Read the existing-ground long section along any alignment found in the DXF, with grade breaks, grade percentages, a regular chainage ladder and a manually set datum line.
- Lay out a proposed driveway profile, including a kerb and a layback template, and check it against configurable AS 2890.1 grade and transition limits.
- Test whether a B85 or B99 design vehicle would scrape along the proposed profile.
- Compare the driveway levels of two alignments (crossfall).
- Save a complete job and reopen it later; export to DXF, CSV and PDF.

1.3 What GradeLine is not

GradeLine is not a general CAD or road-design package. The surface is built from survey points alone, without breaklines. The driveway tool is a screening aid over a design the user authors. Other proposed-surface work remains in the user's CAD or design software.

2. Installation and launch

GradeLine is available in four forms. All four are the same application, and a saved job opens in any of them.

Form	How to obtain and start it	Notes
Windows desktop app	Download **GradeLine_x.y.z_x64-setup.exe** from the latest release on the repository and run it.	Installs for the current user only; no administrator rights are needed. The installer is not code-signed, so Windows may display "Windows protected your PC": choose **More info**, then **Run anyway**.
Browser (installable web app)	Open the hosted application address. In Chrome or Edge, use the browser's install option.	Once installed it works offline and can open **.gradeline** job files.

Form	How to obtain and start it	Notes
Single HTML file	Download **tinviewer.html** from the repository and open it in a browser.	One self-contained file; no server or network access is needed.
Windows installer package	Unzip **GradeLine-installer.zip** and run **Install GradeLine.bat**.	Adds Desktop and Start Menu shortcuts that open the single-file build. No administrator rights are needed.

2.1 First run

1. Open GradeLine. The sidebar is on the left and the main view is on the right (Figure 1).
2. Select **Load sample terrain** to load a built-in demonstration surface.
3. Drag in the 3D view to orbit and scroll to zoom. Hold Shift and drag to pan.
4. Select the **Existing terrain** tab to see the long section along the sample alignment.
5. To work with real data, select **Clear loaded data**, then **Upload DXF** (Section 3).

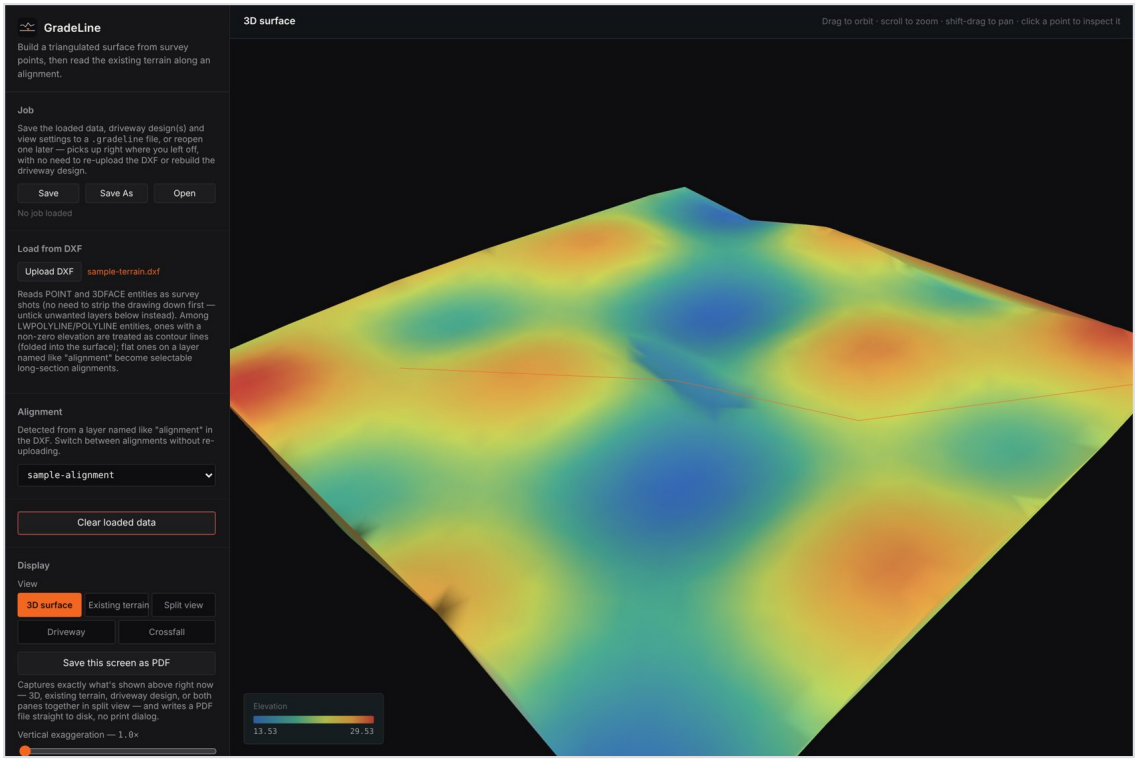


Figure 1. The GradeLine interface with the sample terrain loaded. The sidebar holds the job, DXF, alignment and display controls; the view tabs (3D surface, Existing terrain, Split view, Driveway, Crossfall) select what is shown on the right.

3. Loading survey data

DXF is the only survey format GradeLine reads. Export the survey from the CAD or survey package as a DXF, then select **Upload DXF** in the **Load from DXF** section of the sidebar. There is no need to strip the drawing down first; unwanted layers can be excluded after loading.

3.1 Entities read

DXF entity	Use in GradeLine
`POINT`	Treated as a survey shot (X, Y, Z) and used to build the surface.
`3DFACE`	Face corners are used as survey shots. Corners shared between faces are counted once.
Polyline with a non-zero elevation	Treated as a contour line; its vertices are folded into the surface as additional points.
Flat polyline (no elevation)	Offered as an alignment for the long section.

3.2 Point layers

When the points in a DXF sit on more than one layer, a **Point layers** checklist appears, with a point count beside each layer and **All** and **None** buttons. Only ticked layers feed the surface. With several point layers, none is ticked at first, so that the choice is made deliberately for each load. Tick only genuine survey layers; leave out text labels, blocks and linework.

3.3 Alignments

Flat polylines on a layer whose name contains "alignment" are preferred as alignment candidates. If no such layer exists, every flat polyline is offered. Where there is more than one, the **Alignment** dropdown lists them by layer, point count and length. The selection can be changed at any time without reloading.

After loading, the surface rebuilds automatically. For large files a progress overlay with a **Cancel** button appears; cancelling returns the application to a clean, empty state.

4. Preparing the data

Unsuitable points produce an unreliable surface. Review the data before relying on a section. All filters apply immediately; the file does not need to be reloaded.

4.1 Filters

Control	Default	Purpose
Point layers checklist	Nothing ticked when several point layers exist	Exclude layers that are not survey data.
Ignore points with elevation (Z) = 0	On	Removes DXF text-label rows, which always report Z = 0.000 and would otherwise pull the surface to zero.
Ignore points beyond a coordinate limit	Off (limit 0)	Drops stray outlier clusters, for example points on a mismatched coordinate grid. Set a maximum |X| or |Y| from the survey origin and tick the box.

4.2 Summary and diagnostics

The **Summary** section reports the point count, triangle count, elevation range and alignment length. The **Diagnostics** section is updated after every rebuild and reports:

- **Disconnected clusters**: groups of points that the surface has bridged although they are unrelated, often a stray survey from another site.
- **Near-duplicate points**: points within 0.01 units of each other in X and Y (detection is approximate, using a 0.01 grid).
- **Unusually steep triangles**: possible errors in Z, or genuine cliffs and walls that a triangulated surface cannot represent well.

> **Note** A diagnostic flag is a prompt to inspect the data, not proof of an error. A genuine retaining wall will trigger the steep-triangle check.

Clear loaded data removes all data and returns to the starting state.

5. The views

Five tabs in the **Display** section select the main view. All draw from the same loaded data.

Tab	Content
3D surface	The triangulated ground in 3D, coloured by elevation, with an elevation legend.
Existing terrain	The existing-ground long section along the active alignment.
Split view	The 3D surface stacked above the long section.
Driveway	The long section with the driveway design controls (Section 7).
Crossfall	Two alignments' driveway designs compared (Section 9).

5.1 3D surface

- **Navigation:** drag to orbit, scroll to zoom, Shift-drag to pan.
- **Vertical exaggeration:** a slider from 1x to 10x. It starts at 1x (true scale). Raise it to make gentle terrain readable, and return to 1x before judging real gradients.
- **Display options: Show triangulation edges**, **Show survey points** and **Show alignment on surface**.
- **Point inspection:** click (without dragging) on the surface to read the real-world X, Y and Z of that location in a panel at the lower right.

5.2 Split view

Hovering along the long section places an orange marker at the matching position on the 3D surface, moving smoothly rather than jumping between samples. When driveway design is on, the proposed design line also appears in this view; it is edited on the Driveway tab.

5.3 Saving the current screen as a PDF

Save this screen as PDF captures whichever tab is showing (both panes in split view) and writes a PDF directly to disk, with a header, date and the current alignment name.

6. Reading the long section

The **Existing terrain** tab plots the ground profile along the active alignment, with chainage on the horizontal axis and RL on the vertical axis (Figure 2).

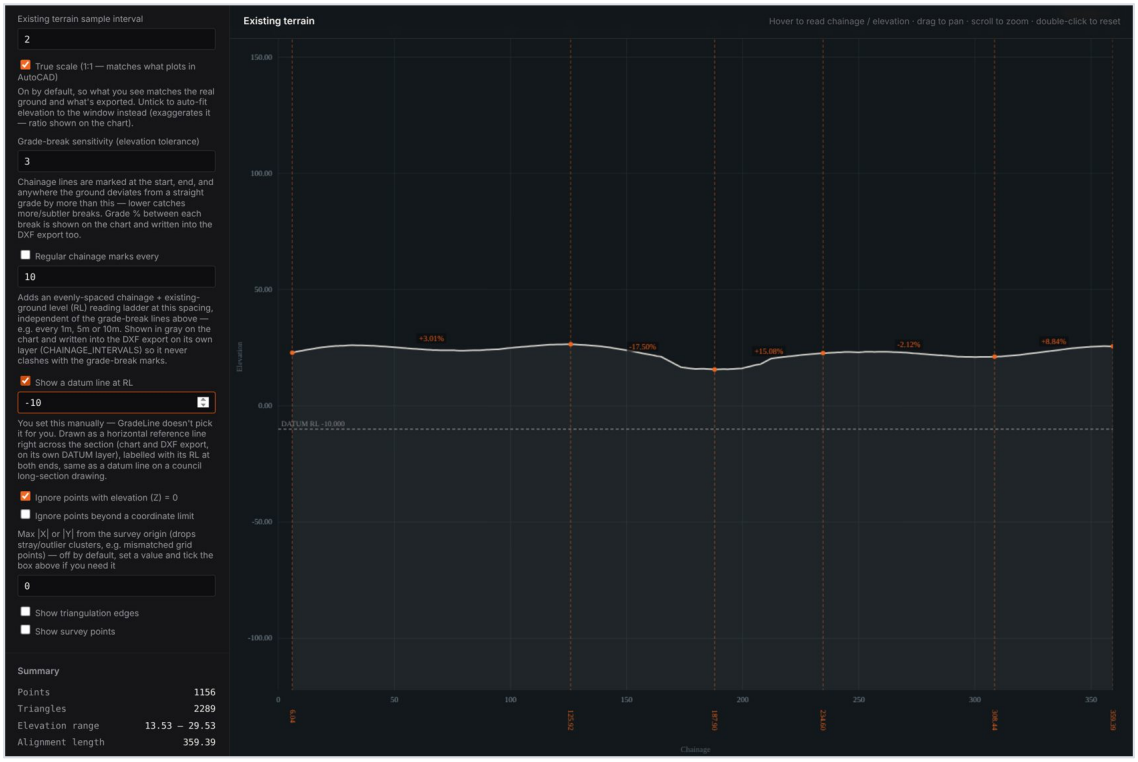


Figure 2. The Existing terrain tab on sample data, showing grade-break chainage lines, signed grade percentages between breaks, a datum line, and the long-section controls in the sidebar.

6.1 Scale and navigation

- **True scale (1:1)** is on by default so the chart matches what plots in AutoCAD. Real sections usually look nearly flat at true scale. Untick it to fit the elevation range to the window; the chart then prints its vertical exaggeration so the distortion is never hidden.
- **Pan and zoom:** drag to pan, scroll to zoom toward the cursor, double-click to reset. Zoom and pan are kept when the sample interval or tab changes, and reset when the data or alignment changes.
- **Hover** over the chart to read chainage and elevation.

6.2 Annotations and controls

Control	Default	Effect
Existing terrain sample interval	2	Adds extra points between the real breaks. It does not change the shape, minimum or maximum of the section, which are found exactly from the surface.
Grade-break sensitivity (elevation tolerance)	0.05	Chainage lines are drawn at the start, the end, and wherever the ground departs from a straight grade by more than this tolerance. A lower value finds more, subtler breaks.
Regular chainage marks every	Off; spacing 10	Adds an evenly spaced chainage and RL reading ladder (grey), independent of the grade breaks.
Show a datum line at RL	Off	Draws a dashed line at an RL entered by the user, labelled "DATUM RL". GradeLine never chooses the datum. It is always exported, even when it lies outside the visible levels.

Between each pair of grade breaks the chart shows the signed grade as a percentage, for example +3.25% or -1.10%. Sample interval, grade-break sensitivity and the chainage ladder are three separate controls: sensitivity decides where grade breaks fall, the ladder uses only the spacing entered, and the sample interval moves no marks.

7. Driveway design

The **Driveway** tab lays a proposed driveway over the existing-ground section and checks it. The model is straight legs between design points, with no vertical curves. It screens a design the user authors; it does not generate one. The chart is at the left and numbered control sections are at the right (Figure 3).

Select **Design a proposed driveway profile** to begin. Every alignment keeps its own independent design: switching the alignment switches which design is edited, and nothing is shared between alignments.

7.1 Kerb and invert (section 1)

Enter the **Kerb invert chainage**, the **Kerb invert RL (gutter level)** and the **Kerb reveal (mm)**, which is the rise from the invert to the top of kerb, then select **Set kerb points**. GradeLine inserts two design points at the same chainage, forming a vertical face. All three fields are deliberately blank by default because kerb details differ between councils. The section can be skipped when the driveway starts flush with the ground.

7.2 Layback and crossover template (section 2)

Two rise-over-run legs, in millimetres, straddle the kerb invert as on a council layback drawing. The road-side leg runs back from the invert, rising away from the gutter low point. The property-side leg runs forward from the top of kerb. The fields are pre-filled with a typical two-segment layback (road side 25 mm rise over 450 mm run; property side 70 mm rise over 450 mm run), which should be checked against the relevant council detail. Select **Apply layback template** to insert the points.

> **Note** The kerb face and both layback legs are never graded or checked against the compliance limits. They are shown as a rise and run in millimetres. This is deliberate and unconditional.

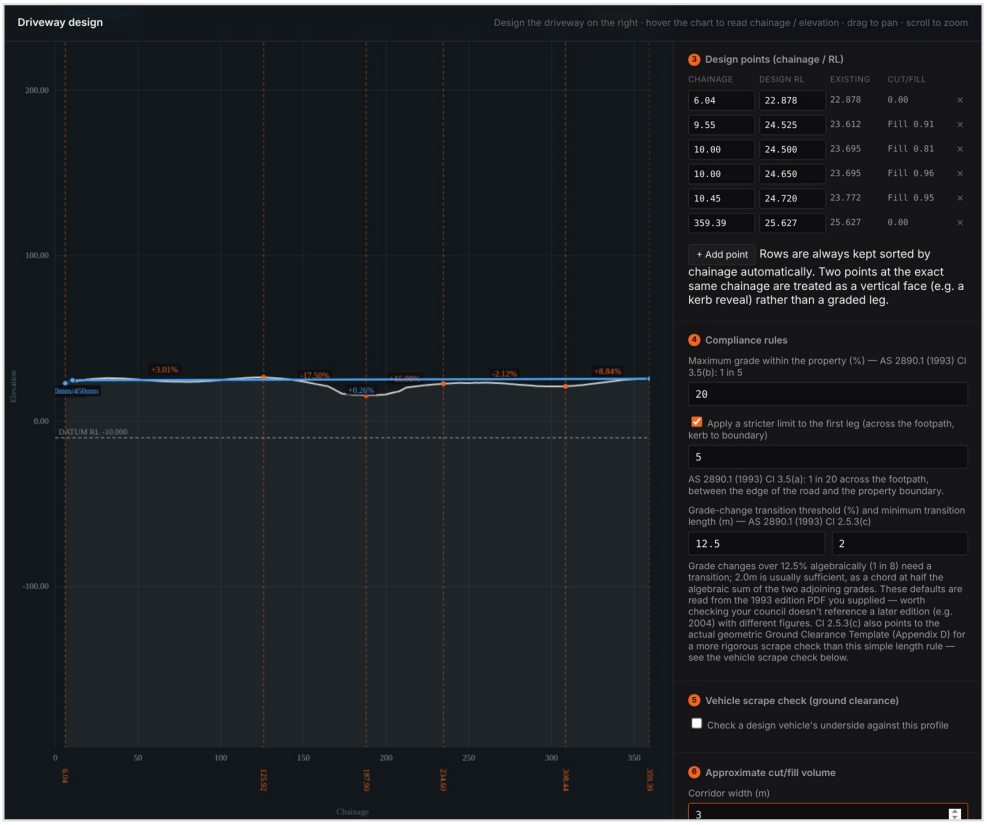


Figure 3. The Driveway tab on sample data after setting kerb points and applying a layback template. Sections 3 (design points) and 4 (compliance rules) are visible at the right. The two rows at chainage 10.00 form a vertical kerb face.

7.3 Design points (section 3)

When design is first enabled, a design point is created at each end of the alignment, matching existing ground. The table lists chainage, design RL, the existing RL and the cut or fill at each point. Values can be edited, points added with **+ Add point** and removed with the cross at the row end. Rows are always sorted by chainage. Two points at exactly the same chainage form a vertical face, shown as a rise in millimetres rather than a grade.

7.4 Compliance rules (section 4)

The thresholds default to the clause text of AS 2890.1-1993 and every value can be changed.

Check	Default	Source
Maximum grade within the property	20% (1 in 5)	Cl 3.5(b)
Stricter limit on the first leg, across the footpath from kerb to boundary	5% (1 in 20), optional	Cl 3.5(a)
Grade change that needs a transition	12.5% (1 in 8)	Cl 2.5.3(c)
Minimum transition length	2.0 m	Cl 2.5.3(c)

The **Compliance summary** lists each leg that passes or exceeds a limit, and failing legs are drawn in red on the chart. The defaults come from the 1993 edition of the Standard. Confirm them against the edition and any local requirements applying to the project.

7.5 Approximate cut and fill (section 6)

Enter a **Corridor width (m)** and GradeLine reports cut, fill and net volume in cubic metres along the design's chainage span. The figure is the corridor width multiplied by the integrated difference between design and existing levels. It is indicative only: it takes no account of crossfall, batters or earthworks shaping, and it is not a surface-to-surface volume.

8. Vehicle scrape check

Section 5 of the Driveway panel tests whether a design vehicle would ground out along the proposed profile. It applies the physical method of AS 2890.1 Appendix D: a rigid underside template is rested on the profile at both wheel contact points and rotated to the local grade. The flat parts of the underside sit at the vehicle's real minimum ground clearance, not flush with the road. Select **Check a design vehicle's underside against this profile** to enable it. Figures 4 and 5 show the check on two example designs.

Control	Function
Design vehicle	B99 or B85. Both use dimensions from 2004 vehicle reference sheets, so treat them as approximations of the published standard, not as its own template.
Vehicle reversing (rear leads)	By default the vehicle faces toward increasing chainage, matching a chart read from street to garage. Ticking this checks the vehicle backing down the driveway. The result can differ, because the front and rear overhangs differ and only the rear has the ramped underside.
Position slider	Sweeps the vehicle along the driveway. It stops when the rear extremity reaches the last design point.
Jump to worst point	Searches the whole design for the tightest clearance and moves the slider there.

Case 1: the vehicle clears. A gently graded driveway falls away from the kerb at 4.75%. The B99 underside, shown as the dashed green line, stays above the profile throughout, and the readout reports **B99 clears by 120 mm at its tightest point**. Because the profile is a single straight leg, the tightest point is simply where the vehicle's own flat clearance sits closest to the ground.

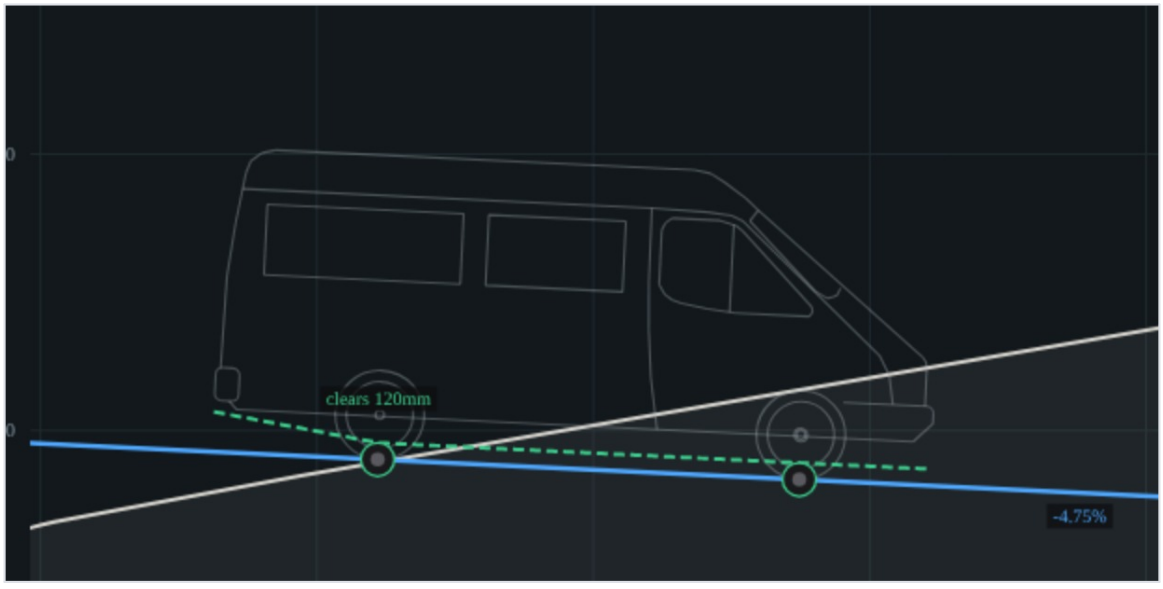


Figure 4. Close-up of the B99 on a 4.75% driveway. The green dashed underside line and wheel markers show that the vehicle clears; the label reads “clears 120 mm”. The solid grey line is existing ground and the blue line is the design.

Case 2: the vehicle scrapes. A driveway that rises to a crest 0.9 m above its two ends (design points at chainages 6.04, 12.00 and 18.00 with levels 24.00, 24.90 and 24.00) is a classic grounding case. With the rear wheel at chainage 10.44 the wheels sit either side of the crest and the vehicle's underside is bridged across it. The dashed line turns red, the label reads **SCRAPES 109 mm**, and the readout gives **B99 SCRAPES by 109mm (worst point at ch 12.00)**.

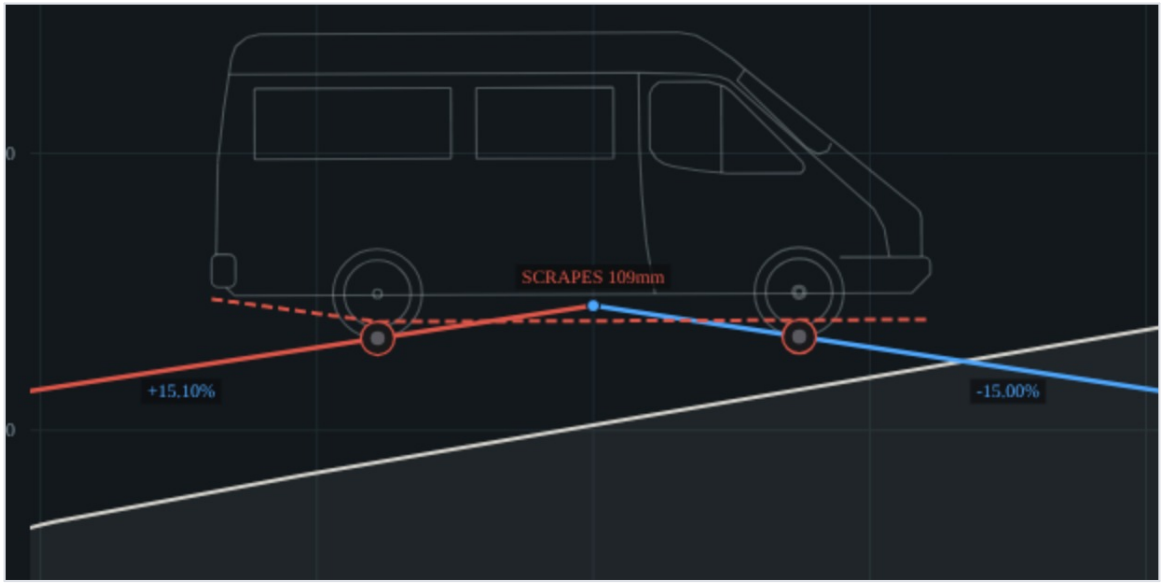


Figure 5. Close-up of the B99 straddling a crest in the design. The red dashed underside line and wheel markers show that the vehicle scrapes at the crest; the label reads “SCRAPES 109 mm”.

The readout reports the clearance, or the amount by which the vehicle scrapes, at the current position. The Compliance summary always includes a worst-case row for the selected direction. On the chart the dashed underside line is green where the vehicle clears and red where it scrapes, with wheel-contact markers and a faint outline of the vehicle drawn beneath the design line.

> **Note** At true 1:1 scale a 100 mm gap is almost invisible beside tens of metres of chainage. Rely on the readout and summary figures, or scroll to zoom in on the vehicle (as in Figures 4 and 5) to inspect a location. This check runs alongside the simpler transition-length rule; the two test different things and can disagree on a marginal case.

9. Crossfall comparison

The **Crossfall** tab compares the driveway levels of two alignments so that the difference in level between them can be read at any chainage. It needs a loaded surface with at least two alignments. Where an alignment has no driveway design enabled, existing ground is used for that side.

9.1 Procedure

1. Select **Alignment A** and **Alignment B**. Two different alignments are chosen by default, and choosing the same alignment twice is refused with a message.
2. Select a **Crossfall comparison method** (Section 9.2).
3. Read the three stacked charts: alignment A's existing ground and design, alignment B's, and a crossfall line (B minus A, in millimetres) against chainage.

All three charts share one true 1:1 scale, so realistic crossfalls look nearly flat. Read the numerical summary rather than the apparent steepness.

9.2 Comparison methods

Method	How points are paired	Suitable when
Matching chainage value (default)	Chainage 10.00 on A is compared with chainage 10.00 on B.	Both alignments start at a comparable point and run roughly parallel.
Perpendicular projection	Each point on A is compared with the nearest point on B's own line in plan.	The alignments start at different places or are not parallel.

Perpendicular mode compares over the whole of alignment A, and the end of B is clamped rather than extended. The default remains matching chainage so that an older saved job never changes method merely by being reopened.

9.3 Reading results

The summary gives the worst crossfall and the chainage at which it occurs, plus a cross-slope percentage once the real horizontal separation between the two lines is known. Drag the orange marker across the chart, or type a chainage in the **Chainage (m)** box, to read the crossfall and each alignment's RL at that point. The marker starts at the worst-crossfall chainage. The Crossfall view has no pan or zoom. It can be exported to DXF (Section 11).

10. Saving and reopening jobs

A job is a **.gradeline** file. It holds the loaded survey data, the chosen alignment, every alignment's driveway design, the crossfall setup, the filter settings, the datum and the view settings. The file is compressed. The surface itself is rebuilt on opening, exactly as after a fresh DXF load.

- **Save**, **Save As** and **Open** (in the **Job** section) use the operating system's own file dialogs, so any folder can be chosen. Save writes back to the last file used; Save As selects a new one.
- If a save fails (for example the drive is disconnected, permission is lost or the disk is full), GradeLine reports it. Use Save As to write elsewhere.
- Browsers without file-picker support (such as Firefox and Safari) download the file on Save and ask for the file on Open.
- In the **desktop app**, double-clicking a **.gradeline** file opens it. The first time, Windows shows a "Pick an app" dialog: choose GradeLine and select **Always**. Saving after a double-click opens Save As.
- In the **installed web app**, double-clicking a **.gradeline** file opens it where the browser supports file handling.

GradeLine also remembers the sidebar's view, filter and datum settings in the browser and restores them at the next visit, even when no job is opened. This never includes survey or design data, and it does nothing quietly if the browser blocks storage.

11. Exporting results

GradeLine exports the long section, the surface and the crossfall view, and can save the current screen as a PDF.

Export (button)	Format	Contents
Download existing terrain (CSV)	CSV	The existing-terrain profile as chainage and elevation.
Existing / Proposed / Both (Export long section)	DXF	Existing ground with grade breaks, grade percentages, the optional chainage ladder and datum; with a driveway design, the proposed design and the section table.
Download TIN surface (DXF, .zip)	DXF	The triangulated ground as `3DFACE` entities in true X, Y, Z.
Existing / Proposed / Both (Export crossfall)	DXF	The three-panel crossfall view with a data table. Proposed and Both are disabled until at least one alignment has a driveway design on.
Save this screen as PDF	PDF	An image of the current tab, written directly to disk.

Some environments that restrict direct file downloads deliver DXF files inside a zip archive; unzip it to obtain the **.dxf**. The desktop app presents a native Save As dialog for every export.

11.1 Contents of the long-section DXF

All geometry is true scale in metres: X is chainage and Y is true elevation, with no vertical exaggeration. Apply the required plot scale in AutoCAD. Each kind of content is on its own layer.

Layer	Contents
`LONG_SECTION`	Existing-ground polyline (white).
`CHAINAGE_MARKS`	Chainage lines and values at the start, the end and each grade break.
`GRADE_LABELS`	Grade percentage between breaks.
`CHAINAGE_INTERVALS`	The regular chainage and RL ladder, if enabled.
`DATUM`	The datum line and its labels, if set.
`PROPOSED_DRIVEWAY`	The design polyline (blue), design-point and leg labels, and any failing checks.
`SECTION_TABLE`	The CAD-style table of distance, grade, existing level, proposed level and chainage.

Labels that would overlap are stacked onto additional rows rather than omitted, and the table columns are spaced apart with leader lines back to their true chainage.

11.2 Opening the DXF in AutoCAD

1. In AutoCAD, choose **File**, then **Open**, set the file type filter to DXF, and select the file.
2. To save a DWG, use **Save As** and choose the DWG file type.
3. To bring the geometry into an existing drawing, open the DXF, copy the geometry, and paste it with **Paste to Original Coordinates**. The **DXFIN** command works only on a blank drawing.

> **Note** GradeLine writes DXF only, in a deliberately plain form with no version header and with the section lines written as old-style polylines. This is the form that opens reliably in AutoCAD. GradeLine cannot write DWG directly, because no free library exists for that format; convert the DXF in AutoCAD or with a converter.

12. Troubleshooting

The following table lists common problems, their likely causes and the action to take.

Symptom	Likely cause and action
The surface has spikes down to zero	Text-label rows with Z = 0 are included. Keep **Ignore points with elevation (Z) = 0** ticked and untick label layers in the Point layers checklist.
A raised bridge or blob far from the site	A stray cluster. Check Diagnostics for disconnected clusters, then untick its layer or set a coordinate limit.
No alignment appears in the dropdown	The DXF contains no flat polyline. Place the alignment on a layer named like "alignment" as a flat polyline and export again.
The long section looks almost flat	This is correct at true 1:1 scale. Zoom in, or untick **True scale** and read the vertical exaggeration label.
The datum line is not visible on the chart	It lies outside the visible levels. It is still written to the DXF.
The vehicle appears to clear but the readout reports a scrape	The gap is too small to see at true scale. Rely on the readout, or zoom in.
Colours or lines are missing when the DXF is opened	Export again from a current version. The ground and design lines are written as old-style polylines so that AutoCAD reads their colours.
Buttons stop responding or the application is slow	Save the job, reload, and reopen the job file.
An export does nothing in the desktop app	Complete the native Save As dialog that appears; check that it is not hidden behind another window.

12.1 Good practice

- Save a **.gradeline** job as soon as the data has been filtered, and again after design changes.
- Review Diagnostics after every load, before relying on a section.
- Judge gradients at true scale and 1x vertical exaggeration; use exaggeration only to see detail.

- Confirm the AS 2890.1 limits against the edition and local requirements that apply to the project before relying on a pass.

13. Limitations and assumptions

- **No breaklines.** The surface is a Delaunay triangulation of the points in plan and does not follow ridges, valleys or kerb lines. A sparse survey across a kerb will smooth over it.
- **Existing ground plus the driveway layer.** There are no full cross-sections and no surface-to-surface earthworks volume. The cut and fill figure is indicative.
- **The driveway design is a screening aid.** It uses straight legs without vertical curves and does not check sight distance. Its compliance defaults come from AS 2890.1-1993 only and do not include any council's own driveway or crossing requirements beyond the kerb and layback shape entered.
- **The kerb face and layback legs are not checked.** This is deliberate.
- **B85 and B99 use reference-sheet dimensions,** not the published Appendix D B99 template. The scrape check and the transition-length rule can disagree on marginal cases.
- **Crossfall** by matching chainage pairs equal chainages and suits alignments that run parallel from a common start. Perpendicular projection handles other cases but remains a straight-line nearest-point comparison.
- **File formats.** GradeLine reads DXF only and writes no DWG.
- **Desktop installer.** It is not code-signed, so Windows SmartScreen may warn on first run, and it does not update itself.

> **Important** GradeLine is an aid to the user's own checking, not a certified design tool. Final design, compliance assessment and sign-off belong in the user's normal design and approval workflow.

Appendix A. Default values

Setting	Default
Vertical exaggeration	1x (true scale)
True scale (1:1)	On
Existing terrain sample interval	2
Grade-break sensitivity (elevation tolerance)	0.05
Regular chainage marks	Off (spacing 10)
Datum line	Off
Ignore points with elevation (Z) = 0	On
Ignore points beyond a coordinate limit	Off (limit 0)
Show triangulation edges / Show survey points	Off / Off
Show alignment on surface	On
Maximum grade within the property	20%
Stricter first-leg (footpath) limit	5%
Grade-change transition threshold / minimum length	12.5% / 2.0 m
Layback template, road side (rise / run)	25 mm / 450 mm
Layback template, property side (rise / run)	70 mm / 450 mm
Design vehicle	B99, facing toward increasing chainage
Crossfall comparison method	Matching chainage value

Appendix B. Glossary

Term	Meaning
Alignment	A plan line along which the long section is cut.
Chainage	Distance along an alignment from its start, in metres.
RL	Reduced level: elevation relative to the survey datum.
TIN	Triangulated irregular network: a surface built from triangles joining survey points.
Long section	A profile of ground (or design) level plotted against chainage.
Grade break	A point where the ground grade changes by more than the sensitivity tolerance.
PVI	Point of vertical intersection: a design point where two straight grades meet.
Kerb invert	The gutter level at the kerb line.
Kerb reveal	The vertical rise from the kerb invert to the top of kerb.
Layback	The lowered kerb profile at a driveway crossover.
Crossfall	The difference in level between two alignments at a comparable position.
Datum line	A horizontal reference line at a chosen RL, drawn across the section.